An adapted version of Q-Methodology provides an in depth understanding of the Impact of climate and health on rural Alaska Native communities

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Introduction

- Indigenous peoples in Alaska stand among our nation's populations most affected by social determinants of health and environmental stressors.
- With the aims to promote health during times of increased rate of environmental shifts, the Alaska Alliance for Community Engagement- Partnership for Action Toward Health (AK ACE- PATH) was developed.
- The overall goals in establishing the AK ACE-PATH was to:
  - Identify highest priority areas in climate and health for rural Alaska Native (AN) communities in the Interior and the Yukon-Kuskokwim Delta.
  - Implement strategies and solutions to mitigate the impacts of local stressors on community well-being.



Figure 1: The map shows the 5 communities in Alaska whrere the AK ACE -PATH team carries out research activities with community partners.

Methodology

We carried out Q-methodology in four rural Alaska Native communities: Arctic Village, Tanana, Toksook Bay, and Emmonak **(figure 1)** to systematically identify complex perspectives related to climate and health priorities. The mixed qualitative-quantitative approach helped to engage the community prioritize the most locally relevant issues, and identify targets for future interventions. We did this following the three steps below in each community.

- First, we conducted focus groups in four rural Alaska Native communities recruiting for broad representation across all community sectors and demographics to identify collective prioritization of climate and health related concerns.
- After obtaining statements from focus groups, we carried out a community-based Q-methodology process in each respective community engaging participants to elicit rankings for climate and health priorities (analyzing participants' rankings together provides a picture of the most important and least important issues among a collective).
- Rankings were documented and entered into an open-source software designed for Q-methodology studies (called EQ Web Sort).

Results

- To assess similarities and differences between communities, we assigned codes to statements from each community. Three coders developed 17 codes, and we assigned up to four codes to each statement, as many statements touched on multiple topics.
- After discussions and iterative coding, we calculated an interrater reliability kappa to achieve a minimum 80% agreement across statements. When comparing the codes assigned to all statements, the most frequently assigned codes are in **table 1 below.**
- Cultural and contextual factors may influence the generation of priorities, as indicated by results from the total codes applied, but when we focused the analysis on the top five highest-ranked priority statements from each community, the most commonly assigned codes to this subset of statements more closely aligned and are listed in **table 2 below.**

Code	# Assigned
Lack of Traditional Foods / Food Security	29
Travel	28
Changes to Plants/Animal Populations and Behaviors	27
Personal/Community Safety Concerns	23
Changes in Weather Conditions (including Extreme Weather Events)	23

Table 1: Most frequently assigned codes in all communities with the number of statements they were assigned to.

Code	# Assigned in top 5
Lack of Traditional Foods / Food Security	7
Loss of Traditional and Cultural Practices (including knowledge transfer)	6
Physical Health	5
Impacts on Young People	4
Animal Health	4

Table 2: Codes assigned to top five ranked statements in each community, and the number of statements among the top 5 a code was assigned to.

Conclusion

- All communities prioritized the transmission of traditional knowledge and practices to the younger generations, the continuity of subsistence practices and the health and accessibility of subsistence resources.
- These priorities indicate the importance of climate and health solutions that increase opportunities for intergenerational knowledge transmission and subsistence engagement in the everyday context of rural AN communities.



Figure 2: Completed Q-Sort Pyramid.



Figure 3: Participant completing Q-Sort pyramid with AK ACE- PATH Co-I, Morag Clinton

Next Steps

Phase II aims to co-produce and test an intervention, Tools for Arctic Health Promotion (TAHP), that targets local environmental stressors and community-level strengths through the engagement of scientific and traditional knowledge in activities delivered to young people and adults in five rural Alaska Native communities.

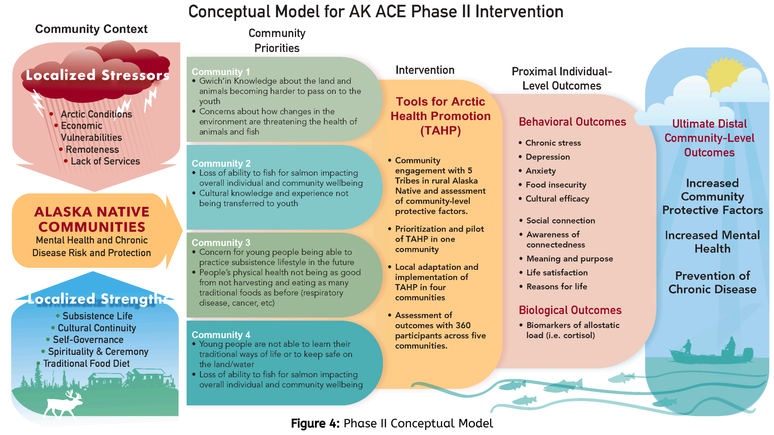


Figure 4: Phase II Conceptual Model

Key Sources & Acknowledgements

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